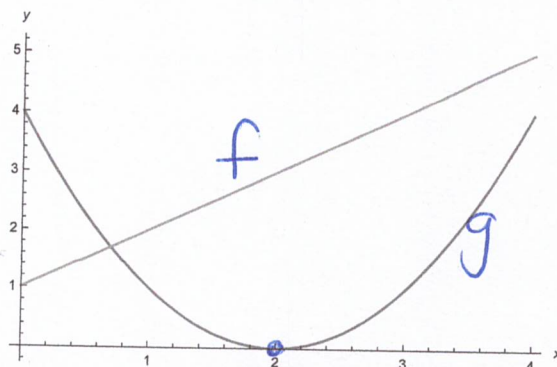


SECTION A [Multiple choice questions] - 18 marks

QUESTION 1

MEMOConsider the graphs of a straight line f and a parabola g below.

$$\begin{aligned}(f \circ g)(2) &= f(g(2)) \\ &= f(0) \\ &= 4\end{aligned}$$

$(f \circ g)(2) =$

(a) 0	☒ (b) 1	(c) 2	(d) 3	(e) 4	(f) 5	(g) None of (a) - (f)
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QUESTION 2

If $\frac{(e^{3x})^2 e^{2x}}{e^{5x}} = e^a$, then $a =$

$$\frac{e^{6x} \cdot e^{2x}}{e^{5x}} = \frac{e^{8x}}{e^{5x}} = e^{8x-5x} = e^{3x} = e^a \Rightarrow a = 3x.$$

☒ (a) $3x$	(b) $-3x$	(c) $9x^2$	(d) $9x^2 - 3x$	(e) $\frac{18}{5}x^2$	(f) None of (a) - (e)
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QUESTION 3

If $5e^{x+\ln 2} = 2$, then $x =$

$$\begin{aligned}e^{x+\ln 2} &= \frac{2}{5} \Rightarrow x + \ln 2 = \ln\left(\frac{2}{5}\right) = \ln 2 - \ln 5 \\ \Rightarrow x &= -\ln 5.\end{aligned}$$

(a) $-\ln 2$	(b) $\ln 2$	☒ (c) $-\ln 5$	(d) $\ln 5$	(e) $-\ln \frac{2}{5}$	(f) $\ln \frac{2}{5}$	(g) None of (a) - (f)
--------------	-------------	---	-------------	------------------------	-----------------------	-----------------------

QUESTION 4

If $f(x) = 2e^{-x}$, then $f^{-1}(x) =$

$$\begin{aligned}y &= 2e^{-x} \\ \text{Let } x &= 2e^{-y} \\ \Rightarrow \frac{x}{2} &= e^{-y} \text{ and } \ln\left(\frac{x}{2}\right) = -y \Rightarrow y = f^{-1}(x) = -\ln \frac{x}{2} = \ln \frac{2}{x}\end{aligned}$$

(a) $\ln \frac{x}{2}$	☒ (b) $\ln \frac{2}{x}$	(c) $\frac{1}{2} \ln x$	(d) $\sqrt{\ln x}$	(e) $\ln(2-x)$	(f) $\ln(x-2)$	(g) None of (a) - (f)
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QUESTION 5

For which value(s) of x is $y = \sin^{-1}(2x-1)$ defined?

(a) $x \in [-1, 1]$	(b) $x \in (-1, 1)$	(c) $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$	(d) $x \in [-\frac{1}{2}, 1]$	☒ (e) $x \in [0, 1]$	(f) $x \in \mathbb{R}$
---------------------	---------------------	---	-------------------------------	---	------------------------

$$\begin{aligned}\sin y &= 2x-1, \quad y \in [-\frac{\pi}{2}, \frac{\pi}{2}] \\ \Rightarrow -1 &\leq 2x-1 \leq 1 \\ \Rightarrow 0 &\leq 2x \leq 2 \\ \Rightarrow 0 &\leq x \leq 1.\end{aligned}$$

QUESTION 6

The relationship between x and y is given by $y = 10 - 2x$. What interval on the y -axis corresponds to the interval $x \in [-40, 100]$ on the x -axis?

(a) $y \in [-190, -90]$	(b) $y \in [-90, 190]$	(c) $y \in [90, 190]$	☒ (d) $y \in [-190, 90]$	(e) None of (a) - (d)
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QUESTION 7

$$\cos \frac{7\pi}{3} + \tan \frac{3\pi}{4} =$$

☒ (a) $-\frac{1}{2}$	(b) $\frac{3}{2}$	(c) $\frac{\sqrt{3}-1}{2}$	(d) $\frac{\sqrt{3}+\sqrt{2}}{2}$	(e) $\frac{\sqrt{3}-\sqrt{2}}{2}$	(f) None of (a) - (f)
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ROUGH WORKING

Q6

$$-40 \leq x \leq 100$$

$$\Rightarrow -80 \leq 2x \leq 200$$

$$\Rightarrow -200 \leq -2x \leq 80$$

$$\Rightarrow 10 - 200 \leq 10 - 2x \leq 10 + 80$$

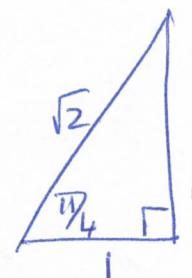
$$\Rightarrow -190 \leq 10 - 2x \leq 90$$

$$\Rightarrow \underline{-190 \leq y \leq 90}$$

Q7 $\cos \frac{7\pi}{3} = \cos (2\pi + \frac{\pi}{3}) = \cos \frac{\pi}{3} = \frac{1}{2}$

$$\tan \frac{3\pi}{4} = \tan (\pi - \frac{\pi}{4}) = -\tan \frac{\pi}{4} = -1.$$

$$\text{and } \cos \frac{7\pi}{3} + \tan \frac{3\pi}{4} = \frac{1}{2} - 1 = -\frac{1}{2}.$$



QUESTION 8

The domain of the function $f(x) = \frac{1}{1 + \ln(x+1)}$ is

- (a) $(-1 + \frac{1}{e}, \infty)$.
- (b) $(-1, \infty)$.
- ☒ (c) $(-1, -1 + \frac{1}{e}) \cup (-1 + \frac{1}{e}, \infty)$.
- (d) $(-1, -1 + \frac{1}{e})$.
- (e) $x \in \mathbb{R}, x \neq -1 + \frac{1}{e}$.
- (f) None of these.

$$\begin{aligned} \textcircled{1} \quad x+1 > 0 &\Rightarrow x > -1 \\ \textcircled{2} \quad 1 + \ln(x+1) &\neq 0 \\ &\Rightarrow \ln(x+1) \neq -1 \\ &\Rightarrow e^{\ln(x+1)} \neq e^{-1} \\ &\Rightarrow x+1 \neq e^{-1} \\ &\Rightarrow x \neq e^{-1} - 1. \end{aligned}$$

QUESTION 9

Consider $F(x) = \sec^4(x-4)$. If $F(x) = (f \circ g \circ h)(x)$, then the functions f, g and h can be represented as follows:

- (a) $f(x) = x - 4, g(x) = \sec x, h(x) = x^4$.
- (b) $f(x) = x - 4, g(x) = x^4, h(x) = \sec x$.
- (c) $f(x) = \sec x, g(x) = x - 4, h(x) = x^4$.
- (d) $f(x) = \sec x, g(x) = x^4, h(x) = x - 4$.
- ☒ (e) $f(x) = x^4, g(x) = \sec x, h(x) = x - 4$.
- (f) $f(x) = x^4, g(x) = x - 4, h(x) = \sec x$.

QUESTION 10

Consider the following statements:

- (1) $(\ln e)^3 = 3$
- (2) $\ln a^2 b^2 = 4 \ln a \ln b$.
- (3) $\ln(1 + e^x) = x$.

$$\begin{aligned} \textcircled{2} \quad \ln a^2 b^2 &= \ln a^2 + \ln b^2 \\ &= 2 \ln a + 2 \ln b \\ \textcircled{1} \quad (\ln e)^3 &= 1^3 = 1. \end{aligned}$$

Choose the correct answer:

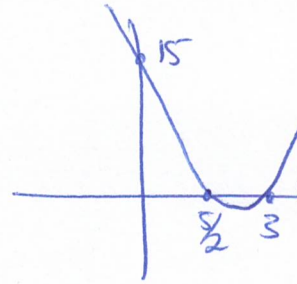
- (a) Only statement (1) is false.
- (b) Only statement (2) is false.
- (c) Only statement (3) is false.
- (d) Only statements (1) and (2) are false.
- (e) Only statements (1) and (3) are false.
- (f) Only statements (2) and (3) are false.
- ☒ (g) Statements (1), (2) and (3) are false.
- (h) Statements (1), (2) and (3) are correct.

QUESTION 11

Solve for x if $\frac{1}{3-x} \geq 2$.

- (a) $x \in (-\infty, 3) \cup (3, \infty)$.
 (b) $x \in (-\infty, \frac{5}{2}] \cup (3, \infty)$.
 (c) $x \in (-\infty, \frac{5}{2}] \cup [3, \infty)$.
 (d) $x \in [\frac{5}{2}, 3)$.
 (e) $x \in [\frac{5}{2}, 3]$.
 (f) None of these.

$$\begin{aligned} \frac{1}{3-x} - 2 &\geq 0 \\ \Rightarrow \frac{1}{3-x} - \frac{2(3-x)}{3-x} &\geq 0 \\ \Rightarrow \frac{1-6+2x}{3-x} &\geq 0 \\ \Rightarrow \frac{2x-5}{3-x} &\geq 0 \\ \Rightarrow (2x-5)(3-x) &\geq 0, x \neq 3 \\ \Rightarrow (2x-5)(x-3) &\leq 0, x \neq 3. \end{aligned}$$



QUESTION 12

The equation of the graph obtained by reflecting the graph of $y = |x-1|$ about the y -axis, stretching it vertically by a factor of 2 and translating it down by 3, is given by:

- (a) $y = 2|x-1| - 3$.
 (b) $y = 2|x+1| - 3$.
 (c) $y = -2|x-1| - 3$.
 (d) $y = 2(|x+1| - 3)$.
 (e) $y = -2(|x-1| - 3)$.
 (f) None of these.

We start with $y = |x-1|$
 Reflected about the y -axis, gives us
 $y = |-x-1|$
 $= |x+1|$

Stretch vertically: $y = 2|x+1|$

End of Section A

ROUGH WORKING

Then: Shift down by 3:
 $y = 2|x+1| - 3$

SECTION B [Fill in questions] - 10 marks

Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 13 [1 mark]

line 1

line 2

Determine the value of a if the line passing through $(a, 1)$ and $(4, 2)$ is parallel to the line passing through $(a+2, 1)$ and $(4, 9)$.

$$m_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-1}{4-a} = \frac{1}{4-a}$$

$$m_2 = \frac{9-1}{4-(a+2)} = \frac{8}{4-a-2} = \frac{8}{2-a}$$

$$a = \frac{30}{7}$$

$$m_1 = m_2 \Rightarrow \frac{1}{4-a} = \frac{8}{2-a}$$

$$\Rightarrow (2-a) = 8(4-a) = 32-8a$$

$$\Rightarrow 7a = 30$$

QUESTION 14 [1 mark]

Assume that $a < 0 < b$. Use interval notation to describe the set in which x lies if x is less than or equal to a or greater than b .

$$x \in (-\infty, a] \cup (b, \infty)$$

QUESTION 15 [1 mark]

If $g(x) = 7 + 2x - e^x$, then $g^{-1}(6) =$

$$p. 66 (17)$$

$$0$$

$$g(0) = 7 + 0 - 1 = 6 \Rightarrow g^{-1}(6) = 0$$

QUESTION 16 [2 marks]

The equation of the function $f(x) = Ca^x$ that passes through $(-1, 3)$ and $(1; \frac{4}{3})$ is

$$p. 54 (22)$$

$$f(x) = 2\left(\frac{2}{3}\right)^x$$

$$f(x) = Ca^x$$

$$\Rightarrow 3 = Ca^{-1} \Rightarrow 3a = C$$

$$\text{Also: } \frac{4}{3} = Ca^1 \Rightarrow \frac{4}{3a} = C$$

$$\Rightarrow 3a = \frac{4}{3a} \Rightarrow 4 = 9a^2$$

$$\Rightarrow a^2 = \frac{4}{9} \text{ and } a = \frac{2}{3} (\text{only})$$

$$3\left(\frac{2}{3}\right) = C = 2$$

QUESTION 17 [2 marks]

- (a) Write down the definition of the function $\cos^{-1} x = \arccos x$.

See p. 64

- (b) Complete the following cancellation property:

$$\cos(\cos^{-1} x) =$$

for $x \in$

p. 64.

QUESTION 18 [1 mark]

Complete the definition of a one-to-one function:
A function $f : A \rightarrow B$ is one-to-one if

Definition 1 on p. 56.

QUESTION 19 [2 marks]

Given $f(x) = \sqrt{x}$, $g(x) = \frac{1}{x^2}$. Determine the following, if defined. Use interval notation, where applicable.

- (a) The domain of $g \circ f$ is

$(0, \infty)$

- (b) $(f \circ g)(x) =$

$\frac{1}{|x|}$

$$\begin{aligned} (g \circ f)(x) &= g(f(x)) \\ &= g(\sqrt{x}), x \geq 0 \\ &= \left(\frac{1}{\sqrt{x}}\right)^2, x > 0 \\ &= \frac{1}{x}, x > 0 \end{aligned}$$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= f\left(\frac{1}{x^2}\right) \\ &= \sqrt{\frac{1}{x^2}} \\ &= \left|\frac{1}{x}\right| \\ &\text{or } \frac{1}{|x|}. \end{aligned}$$

In general:

$$\sqrt{x^2} = +\sqrt{x^2} = |x|.$$

End of Section B

SECTION C [Full written answers] - 22 marks

QUESTION 20 [4 marks]

Consider the function $f(x) = \frac{x-2}{|2-x|}$.

- (a) Simplify f completely, using the definition of the absolute value.

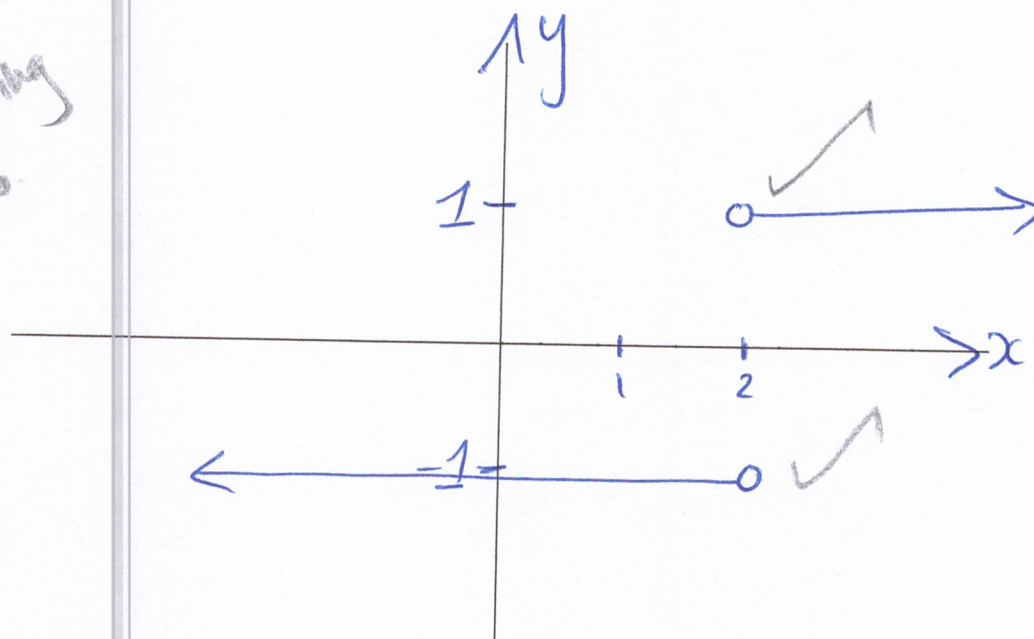
(2)

$$f(x) = \begin{cases} \frac{x-2}{2-x} & \text{if } 2-x > 0, \text{ i.e., } 2 > x \\ \frac{x-2}{-(2-x)} & \text{if } 2-x < 0, \text{ i.e., } 2 < x \end{cases}$$

$$= \begin{cases} -1 & \text{if } x < 2 \\ 1 & \text{if } x > 2 \end{cases}$$

- (b) Sketch the graph of f on the axes provided, indicating all relevant intercepts and information.

(1)



- (c) Write down the domain and range of f .

(1)

$$D_f = (-\infty, 2) \cup (2, \infty) \quad \underline{\text{OR}} \quad \mathbb{R} \setminus \{2\}$$

 $\left(\frac{1}{2}\right)$

$$\underline{\text{Range:}} \quad \{-1, 1\}$$

 $\left(\frac{1}{2}\right)$

QUESTION 21 [8 marks]

Solve for x . Write your answer in interval notation.

(a) $\ln(2x+1) + \ln x > 0$

(3)

$$\Rightarrow \ln(2x+1)x > 0 \quad \text{with} \quad 2x+1 > 0 \quad \text{and} \quad x > 0$$

$$\Rightarrow 2x > -1 \quad \text{and} \quad x > 0$$

$$\Rightarrow x > -\frac{1}{2} \quad \text{and} \quad x > 0$$

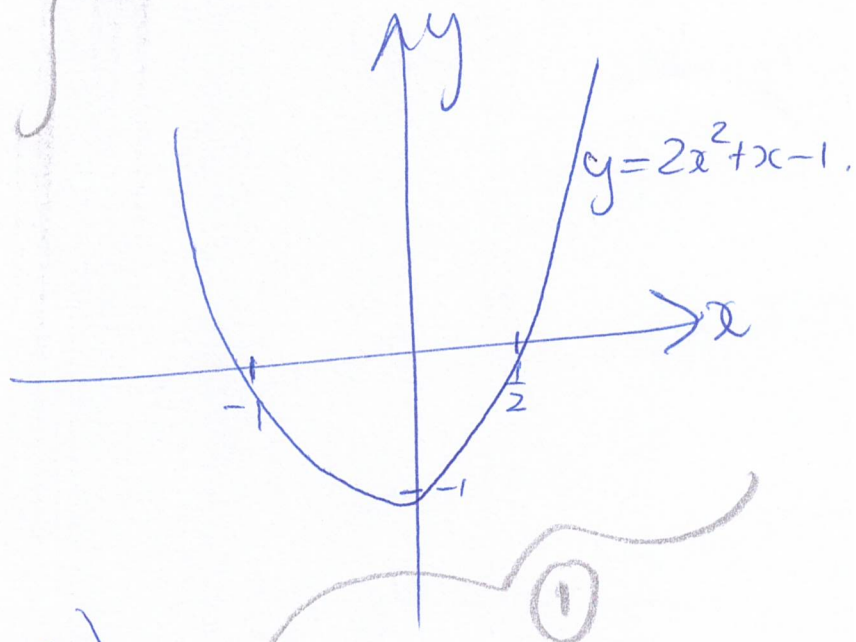
$$\Rightarrow \underline{x > 0}.$$

$$\text{Thus: } e^{\ln(2x+1)x} > e^0$$

$$\Rightarrow (2x+1)x > 1$$

$$\Rightarrow 2x^2 + x - 1 > 0$$

$$\Rightarrow (2x-1)(x+1) > 0$$



$$2x^2 + x - 1 > 0$$

$$\text{if } x \in (-\infty, -1) \cup (\frac{1}{2}, \infty)$$

But, since $x > 0$

$$\text{we have } x \in (\frac{1}{2}, \infty). \quad (1)$$

(b) $e^{bx} > Ce^{ax}$, where $a > b$ and $C > 0$.

$$\begin{aligned} a > b &\Rightarrow a - b > 0 \\ &\Rightarrow b - a < 0 \end{aligned} \quad (3)$$

$$\frac{e^{bx}}{e^{ax}} > C, \text{ since } e^{ax} > 0 \forall x \in \mathbb{R}.$$

$$\Rightarrow e^{bx-ax} > C$$

$$\Rightarrow e^{(b-a)x} > C$$

$$\Rightarrow \ln e^{(b-a)x} > \ln C$$

$$\Rightarrow (b-a)x > \ln C$$

$$\Rightarrow x < \frac{\ln C}{b-a}, \text{ since } b-a < 0$$

(c) $\frac{|x-2|}{x^2+3} > \frac{1}{x^2+3}$

$$\Rightarrow x \in (-\infty, \frac{\ln C}{b-a}).$$

-1 for each mistake.

(2)

$$\frac{|x-2|}{x^2+3} > \frac{1}{x^2+3}$$

$$\Rightarrow |x-2| > 1, \text{ since } x^2+3 > 0 \forall x \in \mathbb{R}. \checkmark$$

$$\Rightarrow \begin{aligned} x-2 &> 1 & \text{or} & x-2 < -1 \\ x &> 3 & \text{or} & x < -1+2 = 1 \end{aligned} \checkmark$$

$$\Rightarrow x > 3 \text{ or } x < 1 \quad \checkmark \quad \textcircled{\frac{1}{2}}$$

$$\Rightarrow \underline{x \in (-\infty, 1) \cup (3, \infty)}. \quad \checkmark \quad \textcircled{\frac{1}{2}}$$

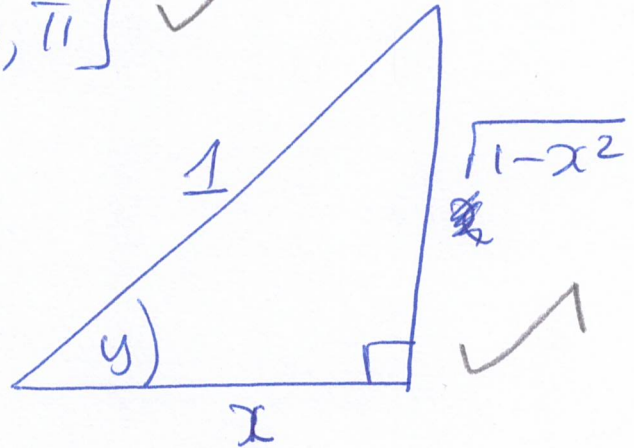
QUESTION 22 [2]

Use the definition in Question 17 and determine the value of

$$\sin(\cos^{-1} x).$$

$$\text{Let } y = \cos^{-1} x.$$

$$\Rightarrow x = \cos y, y \in [0, \pi] \quad \checkmark$$



~~Now $\sin y$~~

$$~~= \sin(\cos^{-1} x)~~$$

$$\text{Now } \sin(\cos^{-1} x)$$

$$= \sin y$$

$$= \sqrt{1-x^2}.$$



QUESTION 23 [3]

Consider the function

$$f(x) = \frac{x+1}{x-2}$$

$$D_f = (-\infty, 2) \cup (2, \infty)$$

Find the inverse of f , together with the domain and range of the inverse function. Use interval notation.

$$y = \frac{x+1}{x-2}$$

$$\text{Let } x = \frac{y+1}{y-2}$$

$$\Rightarrow x(y-2) = y+1$$

$$\Rightarrow xy - 2x = y+1$$

$$\Rightarrow xy - y = 1+2x$$

$$\Rightarrow y(x-1) = 1+2x$$

$$\Rightarrow y = \frac{1+2x}{x-1} = f^{-1}(x)$$

$$D_{f^{-1}} = (-\infty, 1) \cup (1, \infty) = R_f$$

$$R_{f^{-1}} = D_f = (-\infty, 2) \cup (2, \infty)$$

$\left(-\frac{1}{2}\right)$ for
missing/wrong
info

QUESTION 24 [5 marks]

(a) Write down the definition of an odd function.

(1)

See p. 17 of textbook: Stewart (8th ed).

Assume that f and g are odd functions with the same domains. Is the function

$$y = (fg)(x)$$

an odd function? Explain your answer.

(2)

$$\left. \begin{array}{l} f \text{ odd} \Rightarrow f(-x) = -f(x) \\ g \text{ odd} \Rightarrow g(-x) = -g(x) \end{array} \right\} \checkmark$$

Consider $(fg)(-x)$

$$\begin{aligned} (fg)(-x) &= f(-x)g(-x) && \text{(def. } fg) \\ &= -f(x) \times (-g(x)) && \text{(} f, g \text{ odd)} \\ &= f(x)g(x) \\ &= (fg)(x) \neq -(fg)(x). \end{aligned} \quad \left. \right\} \textcircled{1}$$

$\Rightarrow fg$ is not an odd function.

(b) Consider the functions

$$f(x) = a^x, a > 0, \text{ and } g(x) = e^{kx}, k \in \mathbb{R}.$$

Given that $f(x) = g(x)$, in which interval does a lie if $k > 0$.

(2)

$$f(x) = g(x)$$

$$\Rightarrow a^x = e^{kx} \quad \forall x \in \mathbb{R}.$$

$$\Rightarrow \ln a^x = \ln e^{kx}$$

$$\Rightarrow x \ln a = kx \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \ln a = k$$

$$\text{If } k > 0, \text{ then } \ln a > 0$$

$$\Rightarrow e^{\ln a} > e^0$$

$$\Rightarrow a > 1$$

$$\Rightarrow \underline{a \in (1, \infty)}.$$